# Precision CRISPR-Cas9 Gene Editing for Targeted Cancer Immunotherapy

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## ABSTRACT

CRISPR-Cas9 technology has revolutionized genome editing by enabling precise modifications in target DNA sequences. This paper evaluates lipid nanoparticle (LNP) mediated delivery of Cas9 mRNA and sgRNA targeting immunosuppressive checkpoints in chimeric antigen receptor (CAR) T-cells for solid tumor oncology.

## 1. INTRODUCTION

Adoptive cell therapy with CAR T-cells has shown remarkable efficacy in hematologic malignancies, but encounters microenvironmental barriers in solid tumors. Disruption of immune checkpoint genes such as PDCD1 (PD-1) via gene editing restores T-cell effector functions.

## 2. METHODOLOGY & DELIVERY

We engineered electroporation-free LNP formulations encapsulating Cas9 ribonucleoprotein (RNP) complexes. In vitro knock-out efficiency exceeded 92% in primary human T-cells without compromising cell viability.

## 3. RESULTS & CLINICAL IMPLICATIONS

In xenograft mouse models of glioblastoma, CRISPR-edited CAR T-cells demonstrated sustained tumor regression, enhanced cytokine production (IFN-gamma, IL-2), and prolonged persistence compared to unedited controls.

## 4. RESEARCH GAPS & ETHICAL CONSIDERATIONS

Off-target cleavage risks and potential chromosomal translocations require high-throughput unbiased off-target profiling (e.g., GUIDE-seq). Long-term clinical safety monitoring remains essential.

## 5. CONCLUSION

Targeted disruption of inhibitory pathways using LNP-delivered CRISPR-Cas9 enhances anti-tumor activity of CAR T-cells, paving the way for next-generation cell immunotherapies.